YOUTH FOR EARTH 2025

PROJECT REPORT

1. Title of the Project

NAME OF THE TEAM: EcoCampus: Transforming Waste into Green Resources

TEAM MEMBERS:

1. Tanisha Rawat
2. Himadri Rajput
3. Aum Waradkar
4. Shravani Pachpute
5. Jyoti Yadav

Mentor Names - Dr.Anuradha C Phadke
Email of the Mentor – anuradha.phadke@mitwpu.edu.in
Mobile- +91 9423569145

Date of Inception of the Project - 21st April 2025

Date of Completion of the Project – It's a year and half long project and will be completed by the end of 2026.

2. Introduction

In response to the **growing challenges of efficient waste management**, our project introduces a smart, automated waste segregation system powered **by IoT and realtime data analytics**. The system is designed to identify, sort, and monitor mixed waste with minimal human intervention, ensuring faster processing and reducing contamination of recyclables.

Our methodology is divided into three core components:
Hardware Setup – A physical tunnel-based structure that classifies waste using sensors and automated diverters*
System Workflow & Data Collection – An ESP32 microcontroller manages realtime waste classification and bin monitoring, transmitting data to the cloud*
Software & Visualization – A user-friendly web interface displays analytics on waste type, bin levels, and area-wise waste patterns to support informed decisionmaking.

Together, these components form a seamless and scalable solution that promotes sustainability and data-driven waste management for smarter cities.

---

3. Aims

To develop an automated, sensor-based waste segregation system integrated with IoT and data visualization tools, in order to enable efficient, real-time waste classification, monitoring, and data-driven decision-making for sustainable waste management.

Objectives

- Design and implement a tunnel-based hardware system capable of segregating mixed waste into organic and inorganic categories using moisture sensors
- Integrate an ESP32 microcontroller to collect real-time data from moisture and level sensors for continuous monitoring of waste type and bin fill status
- Enable wireless data transmission to the cloud using the ESP32's built-in Wi-Fi for remote tracking and storage of sensor data

- Develop a custom web application to visualize:
  - Waste classification (organic/inorganic)
  - Real-time bin fill level
  - Area-wise waste generation statistics
- Reduce manual intervention and contamination of recyclables through automated segregation
- Promote sustainability by enabling authorities and citizens to make informed, data-backed decisions for effective waste management.

---

4. Methodology

Hardware Setup
Overview of the Physical System:

- The waste segregation system is built on a tunnel-based structure, where mixed garbage (organic & inorganic) is initially collected.
- A moisture sensor is placed at the initial stage of the tunnel to analyze the moisture content of the waste:
  - High moisture content indicates organic waste.
  - Low or no moisture indicates inorganic waste.
- Based on the sensor reading, an automated diverter mechanism directs the waste into one of two separate sub-tunnels:
  - One for organic waste
  - One for inorganic waste
- Each sub-tunnel leads to designated bins for proper storage and monitoring.

System Workflow and Data Collection
Functionality of the IoT-Integrated System:

- The ESP32 microcontroller reads real-time data from:
  - Moisture sensor (for waste classification)
  - Level sensors (to detect how full each waste bin is)
- All sensor data is transmitted to the cloud using the ESP32's built-in Wi-Fi.
- This enables continuous monitoring of:

    - Type of waste being processed
    - Quantity of waste collected in bins
- The system ensures automated and efficient segregation, reducing the need for manual sorting and minimizing contamination of recyclable materials.

Software and Data Visualization
Web Application for Monitoring and Analytics:

- A custom-built website serves as the user interface for interacting with the system.
- The website displays:
    - Type of waste segregated (organic/inorganic)
    - Bin fill levels using level sensor data
    - Area-wise waste statistics, based on the data uploaded to the cloud

Users can monitor:

- Trends in waste generation
- Efficiency of segregation
- Which type of waste is more dominant in specific locations

The system aims to create a data-driven waste management solution, helping both authorities and citizens take informed actions for sustainability.

Figure 1 indicates a complete block diagram visualization for our project

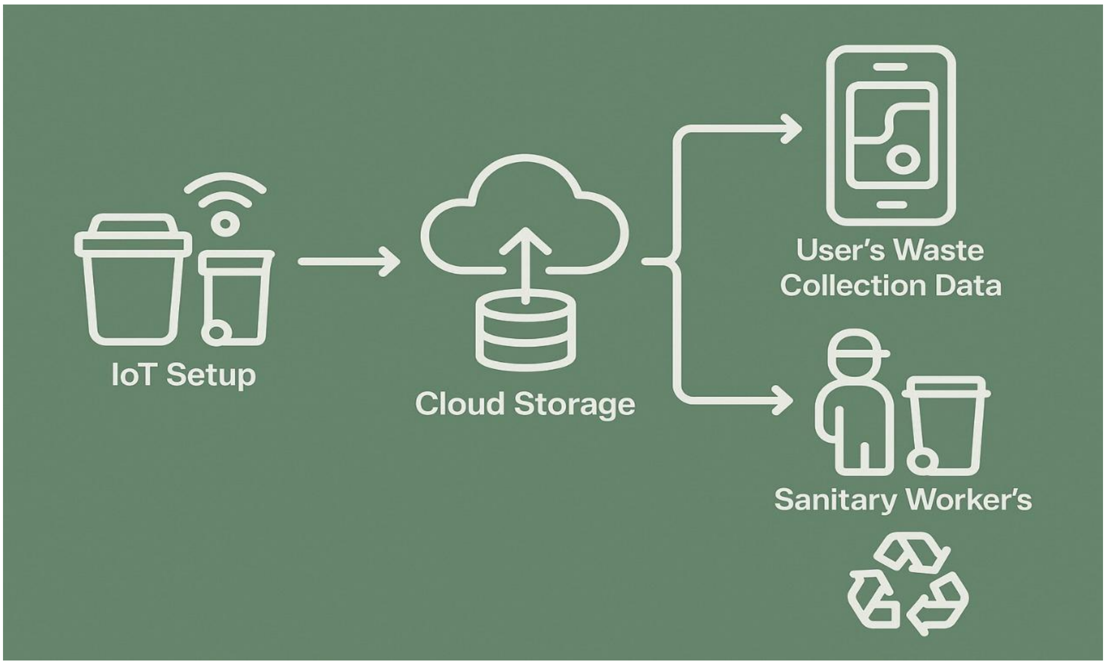

Fig 1- Block Diagram for the project

This is How our app looks like:

Fig 2- Dashboard screen of our app

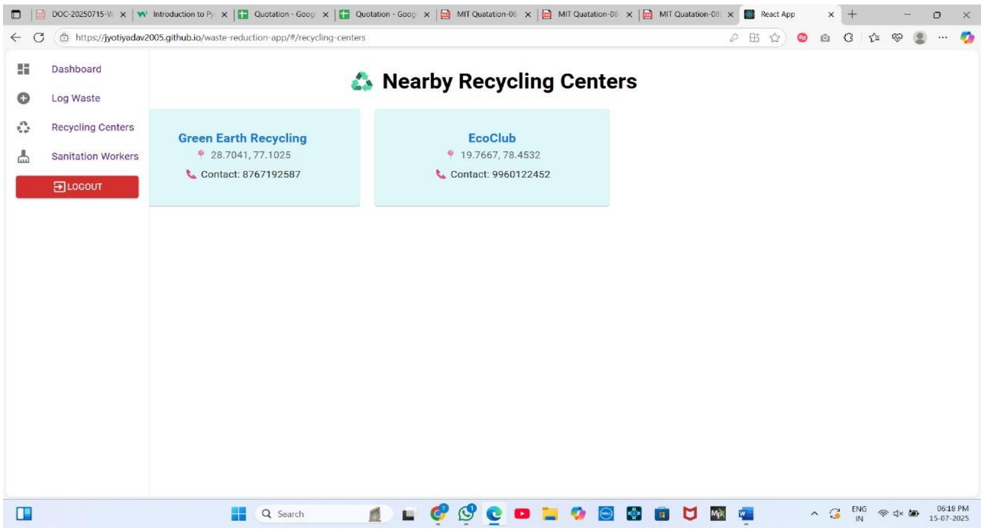

Fig 3- Contact screen

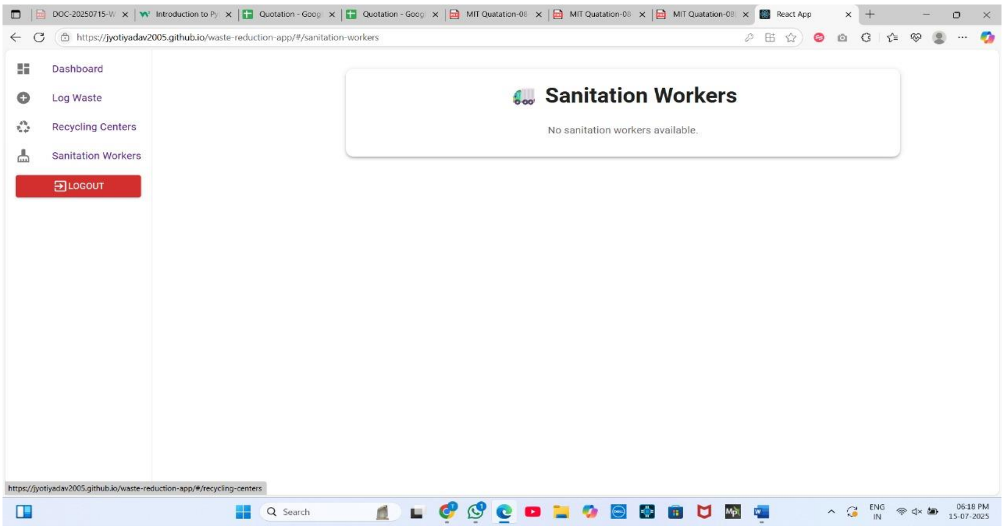

Fig 4-Sanitation Workers contact

About Our App:

- The Waste Reduction App is a clean, minimal web app built to raise awareness and encourage daily waste tracking among households. The goal is to help users become conscious of their waste habits and promote sustainable practices through self-reporting and data visualization.
- Users can segregate waste into- Organic, Plastic or Metal.
- They get reward points according to the amount of waste they log.
- This web app also helps recycling centers to identify high-volume areas to collect waste from which can be recycled or organic waste which can be converted to compost.
- Built Using React and it uses Firebase for the data storage
- This will also help Sanitation Workers track households better and complete their waste collection job more effectively.
- Figures 2, 3 and 4 give a picture of what our app looks like.

5. Environmental & Social Impacts

 Reduces Landfill Waste:
Accurately segregating organic and inorganic waste diverts waste from landfills, reducing soil and groundwater pollution.
 Minimizes Recycling Contamination:
Proper separation at the source reduces contamination, making recycling more effective and sustainable.
 Encourages Composting:
Organic waste segregation supports composting and bio-processing, lowering methane emissions and promoting circular waste management.
 Promotes Eco-conscious Living:
Real-time insights help identify high-waste areas, encouraging community awareness and waste-reducing habits.

---

6. Social Impact

- **Improves Public Health:**
Automated handling reduces human contact with waste, lowering disease risk and improving sanitation.

- **Reduces Manual Labor Burden:**
Minimizes unsafe manual segregation, improving dignity and safety for waste workers.

- **Empowers Citizens & Authorities:**
Transparent data helps communities and policymakers take targeted action.

- **Supports Smart City Goals:**
Merges technology and sustainability to meet urban cleanliness and development goals.

---

7. Challenges Faced in Implementation

- **Sensor Accuracy and Calibration:**
Moisture sensors may give inconsistent readings, requiring frequent calibration.

- **Waste Overlap and Mixed Material Detection:**
  Complex or semi-wet waste (e.g., oily plastics) may lead to misclassification.

- **Hardware Durability:**
  Components may face damage due to moisture, dirt, or heavy waste.

- **Wi-Fi and Cloud Connectivity Issues:**
  Unstable connections can delay or block real-time monitoring.

- **Data Storage and Management:**
  Large-scale data needs secure, scalable cloud infrastructure.

- **User Adoption and Maintenance:**
  Long-term success depends on regular maintenance and user cooperation.

8. Analysis of Waste Segregation in Daily Life

Insights were collected from **various individuals between the age group of 17 to 30 years** via a survey. Their responses helped evaluate the system's perceived effectiveness and practical value.

Figures 5, 6,7,8,9,10 and 11 gives us an idea about waste disposal techniques and awareness in individuals on the basis of the survey taken.

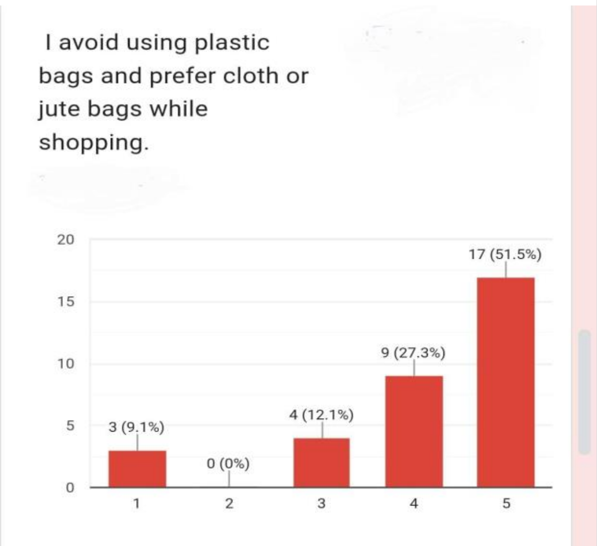

Fig 5

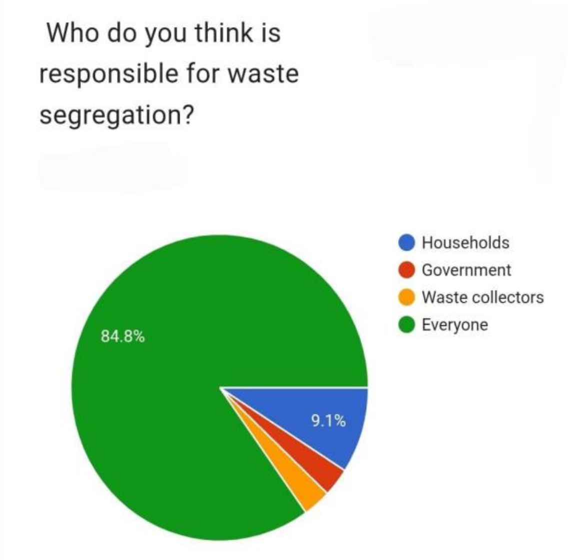

Fig 6

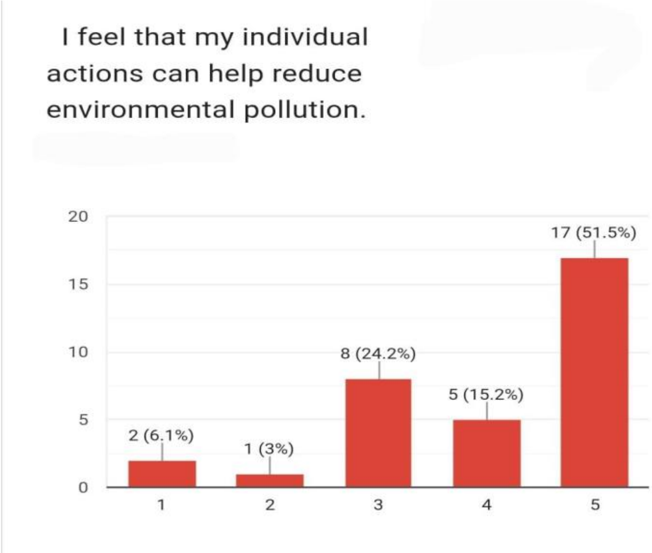

Fig 7

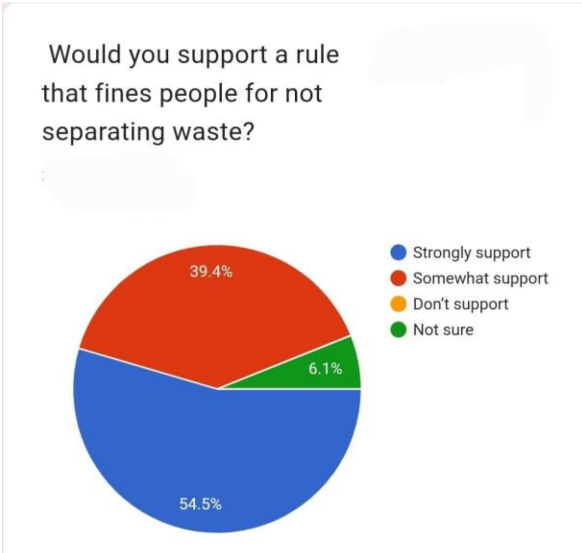

Fig 8

Fig 9

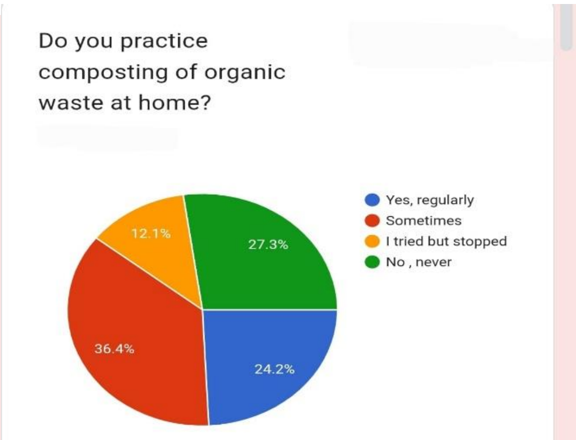

Fig 10

Form_Responses

	Timestamp	How do you usually dispose of your kitchen (	Do you practice composting of organic waste	How do you dispose of plastic waste (bottles	How often do you recycle inorganic wast
2	21/06/2025 10:58:50	Compost at home	Sometimes	Separate and give to recycler	Sometimes
3	21/06/2025 12:22:03	Give to municipality	No , never	Throw with other garbage	Sometimes
4	21/06/2025 12:23:47	Throw with general garbage	No , never	Throw with other garbage	Never
5	21/06/2025 12:29:18	Throw with general garbage	Sometimes	Separate and give to recycler	Always
6	21/06/2025 12:31:36	Give to municipality	Yes, regularly	Separate and give to recycler	Sometimes
7	21/06/2025 12:31:53	Compost at home	Yes, regularly	Separate and give to recycler	Rarely
8	21/06/2025 12:34:32	Throw with general garbage	No , never	Throw with other garbage	Never
9	21/06/2025 12:36:57	Throw with general garbage	Sometimes	Separate and give to recycler	Always
10	21/06/2025 12:39:20	Throw with general garbage	No , never	Throw with other garbage	Never
11	21/06/2025 12:39:41	Throw with general garbage	No , never	Throw with other garbage	Sometimes
12	21/06/2025 12:43:40	Compost at home	Yes, regularly	Throw with other garbage	Rarely
13	21/06/2025 12:48:15	Throw with general garbage	No , never	Separate and give to recycler	Sometimes

Fig 11

---

9. Result

- **Effective Waste Segregation:**
  Successfully classified mixed waste using moisture sensors.

- **Automated Operations:**
  Diverter mechanism minimized manual handling and ensured efficient waste redirection.

- **Real-Time Monitoring:**
  ESP32 successfully transmitted sensor data to the cloud for live tracking.

- **User-Friendly Web Interface:**
  Dashboard displayed waste statistics, trends, and bin status clearly.

- **Data-Driven Insights:**
  Area-wise waste trends supported better waste strategy planning.

- ## Conclusion

The smart waste segregation system proves that IoT, automation, and data visualization can solve environmental challenges. By automating sorting and enabling remote monitoring, the system reduces human effort, enhances recycling, and supports smarter decision-making. Its scalability makes it ideal for smart cities and eco-conscious communities.

- ## Future Potential

- **Multi-Sensor Integration:**
  Add gas sensors -to detect hazardous or toxic gases, temperature sensor-for identifying flammable materials, or RFID sensors for improved safety and classification.

- **AI-Based Waste Recognition:**
  Use machine learning and image recognition to classify complex waste types.

- **Mobile App Integration:**
  Notify users with alerts, stats, and tips to encourage engagement.

- **Predictive Analytics:**
  Forecast waste trends to improve planning and logistics.

- **Collaboration with Municipal Bodies:**
  Scaling up with government and private partnerships to enhance impact.
